

KOŁOBRZEG, Poland

WMO: 121000

Lat: 54.1825N

Lon: 15.5797E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.0	-14.5	1.1	-9.7	-11.7	1.4	-7.4	15.4	4.2	14.0	3.6	2.2	140	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	26.3	19.2	24.2	18.9	22.6	18.2	20.7	24.4	19.7	23.1	18.8	21.7	2.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.1	22.5	18.4	13.3	21.4	17.6	12.7	20.6	59.5	24.6	56.3	23.3	53.5	21.8	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.1	8.3	-12.1	31.5	3.7	2.6	-14.8	33.3	-17.0	34.8	-19.1	36.3	-21.8	38.2		
(5)			-12.7	22.3	3.6	1.0	-15.3	23.0	-17.4	23.6	-19.4	24.1	-22.0	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	0.4	0.9	3.3	7.8	11.9	15.8	17.8	17.9	14.8	10.1	5.8	2.7	(6)
(7)		DBStd	7.08	4.66	3.96	3.73	3.32	3.54	2.93	2.22	2.51	2.48	3.08	2.96	3.94	(7)
(8)		HDD10.0	1252	298	254	208	84	18	1	0	0	1	37	127	225	(8)
(9)		HDD18.3	3414	556	488	465	317	202	88	37	38	109	256	376	483	(9)
(10)		CDD10.0	941	0	0	2	17	78	173	242	245	144	39	2	0	(10)
(11)		CDD18.3	62	0	0	0	0	3	11	21	24	3	0	0	0	(11)
(12)		CDH23.3	295	0	0	0	2	39	76	60	106	12	0	0	0	(12)
(13)		CDH26.7	71	0	0	0	0	7	23	8	32	1	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.8	3.7	3.6	3.5	3.3	3.3	3.2	3.0	3.1	3.2	3.5	4.0	(14)
(15)	Precipitation	PrecAvg	707	49	40	45	35	54	72	77	89	72	60	56	57	(15)
(16)		PrecMax	915	125	68	99	84	101	193	164	201	153	149	110	107	(16)
(17)		PrecMin	584	2	5	9	8	13	8	4	11	17	19	5	17	(17)
(18)		PrecStd	87	28	16	23	19	21	35	41	41	32	34	22	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	11.1	15.9	21.8	27.8	29.5	27.7	30.5	25.2	19.9	13.5	11.7	(19)
(20)			MCWB	9.4	8.4	9.9	13.7	18.2	19.9	20.4	20.4	17.4	14.7	11.8	10.0	(20)
(21)		2%	DB	8.9	8.9	12.0	17.7	23.3	25.1	24.7	25.6	22.6	17.7	12.1	10.1	(21)
(22)			MCWB	7.5	6.8	9.2	11.1	16.4	18.8	19.9	19.8	16.7	14.4	10.4	8.7	(22)
(23)		5%	DB	7.3	7.3	10.0	14.8	19.9	22.2	23.3	23.5	20.5	16.1	11.1	8.6	(23)
(24)			MCWB	6.1	5.6	7.5	10.2	14.9	17.6	19.4	19.0	16.4	13.5	9.7	7.2	(24)
(25)		10%	DB	6.1	5.8	8.3	12.7	17.3	20.0	21.9	21.9	18.9	14.6	10.2	7.3	(25)
(26)			MCWB	4.9	4.4	6.3	9.2	13.7	16.7	18.6	18.2	15.6	12.7	9.0	6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	9.3	11.4	13.8	18.7	20.7	22.1	22.0	19.4	15.9	12.4	10.4	(27)
(28)			MCDWB	11.2	10.0	14.2	19.1	25.1	26.2	25.8	27.1	22.4	18.2	13.2	11.4	(28)
(29)		2%	WB	7.6	7.0	9.3	11.9	17.1	19.3	20.8	20.6	17.7	14.8	10.8	8.9	(29)
(30)			MCDWB	8.8	8.6	11.5	16.2	22.7	24.7	23.7	24.5	20.5	17.1	11.7	10.0	(30)
(31)		5%	WB	6.3	5.7	7.7	10.7	15.5	18.0	19.8	19.6	16.9	13.8	9.9	7.4	(31)
(32)			MCDWB	7.3	7.1	9.7	14.3	19.2	21.3	22.5	22.8	19.5	15.6	10.9	8.5	(32)
(33)		10%	WB	5.0	4.5	6.4	9.5	14.0	17.0	18.9	18.7	16.1	12.8	9.0	6.2	(33)
(34)			MCDWB	6.0	5.7	8.0	12.5	17.0	19.8	21.2	21.4	18.5	14.4	10.1	7.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.7	4.1	5.3	6.2	6.7	6.5	6.1	7.0	6.7	5.7	3.9	3.5	(35)
(36)			MCDBR	3.9	5.9	8.2	10.7	12.4	10.9	9.3	9.9	9.7	7.8	4.7	4.5	(36)
(37)			MCWBR	3.7	4.4	5.3	6.4	6.8	5.9	5.4	5.4	5.5	5.2	3.8	4.0	(37)
(38)		5% WB	MCDWR	3.9	5.5	7.3	9.4	10.9	9.3	8.1	8.7	7.8	7.0	4.4	4.4	(38)
(39)			MCWBR	3.7	4.5	5.1	6.0	6.5	5.7	5.2	5.4	5.3	5.2	3.9	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.342	0.369	0.406	0.405	0.399	0.423	0.421	0.383	0.364	0.343	0.314	(40)	
(41)		taud	2.304	2.308	2.274	2.239	2.307	2.352	2.299	2.313	2.395	2.390	2.343	2.307	(41)	
(42)		Ebn at Noon	627	739	800	818	842	850	819	796	782	706	589	549	(42)	
(43)		Edh at Noon	59	81	103	122	120	116	121	113	92	74	57	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.49	1.13	2.40	4.22	5.42	5.75	5.26	4.35	3.14	1.58	0.63	0.34	(44)	
(45)		RadStd	0.06	0.17	0.23	0.50	0.61	0.41	0.62	0.36	0.32	0.23	0.09	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-132	N/A	N/A

Nomenclature: See separate page